

Practice

Tuesday, 8 September 2026 10:01 am

Model the "reorder past meal" process as a BPMN flowchart.

Write 5 SQL SELECT queries against a sample database (use a free sample DB like Northwind) to answer business questions like "top 5 customers by order value."



1. Top 5 customers by order value

```
SELECT TOP (5)
    c.CompanyName,
    SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalSales
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
JOIN [Order Details] od ON o.OrderID = od.OrderID
GROUP BY c.CompanyName
ORDER BY TotalSales DESC;
```

	AZ CompanyName	123 TotalSales
1	QUICK-Stop	110,277.304977417
2	Ernst Handel	104,874.9787139893
3	Save-a-lot Markets	104,361.9499206543
4	Rattlesnake Canyon Grocery	51,097.8003330231
5	Hungry Owl All-Night Grocers	49,979.9050006866

2. Top 5 best-selling products

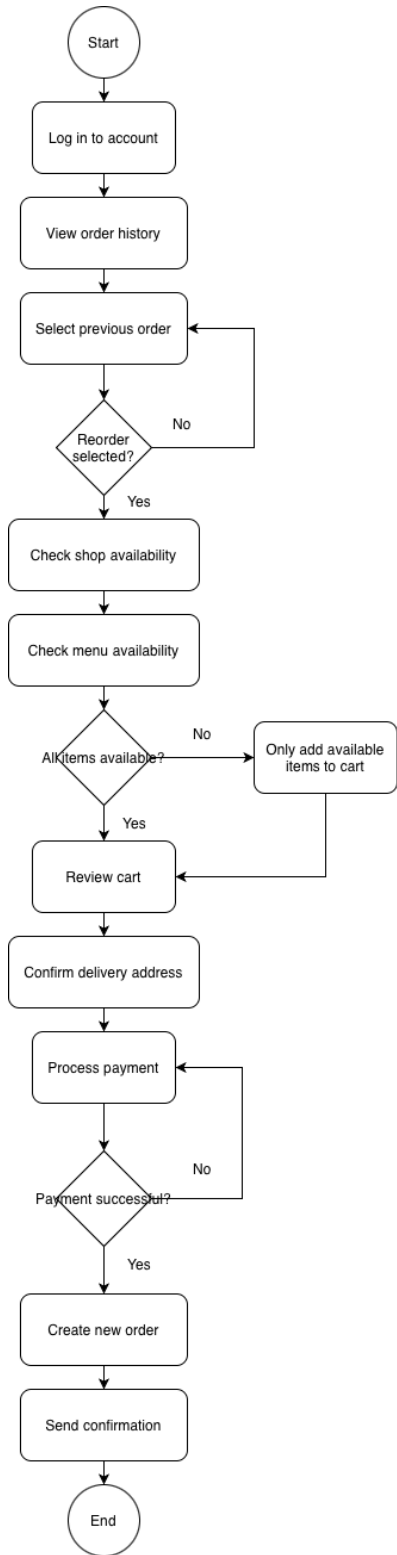
```
SELECT TOP (5)
    p.ProductName,
    SUM(od.Quantity) AS TotalUnitsSold
FROM Products p
JOIN [Order Details] od ON p.ProductID = od.ProductID
GROUP BY p.ProductName
ORDER BY TotalUnitsSold DESC;
```

	AZ ProductName	123 TotalUnitsSold
1	Camembert Pierrot	1,577
2	Raclette Courdavault	1,496
3	Gorgonzola Telino	1,397
4	Gnocchi di nonna Alice	1,263
5	Pavlova	1,158

3. Sales by employee

```
SELECT
    e.FirstName + ' ' + e.LastName AS EmployeeName,
    SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalSales
FROM Employees e
JOIN Orders o ON e.EmployeeID = o.EmployeeID
JOIN [Order Details] od ON o.OrderID = od.OrderID
GROUP BY e.FirstName, e.LastName
ORDER BY TotalSales DESC;
```

	AZ EmployeeName	123 TotalSales
1	Margaret Peacock	232,890.8459472656
2	Janet Leverling	202,812.8427934647
3	Nancy Davolio	192,107.6043205261
4	Andrew Fuller	166,537.7549781799
5	Laura Callahan	126,862.2777042389
6	Robert King	124,568.2353820801



Order	7	Anne Dodsworth	77,308.0667095184	
	8	Michael Suyama	73,913.1292438507	
	9	Steven Buchanan	68,792.2825393677	

4. Which customers place the most orders

```
SELECT TOP (10)
    c.CompanyName,
    COUNT(o.OrderID) AS NumberOfOrders
FROM Customers c
JOIN Orders o ON c.CustomerID = o.CustomerID
GROUP BY c.CompanyName
ORDER BY NumberOfOrders DESC;
```

Customers(+) 1 X

SELECT TOP (10) c.CompanyName, COUNT(o.OrderID) AS NumberOfOrders

Grid	AZ CompanyName	123 NumberOfOrders
Text	1 Save-a-lot Markets	31
	2 Ernst Handel	30
	3 QUICK-Stop	28
	4 Folk och få HB	19
	5 Hungry Owl All-Night Grocers	19
	6 Berglunds snabbköp	18
	7 HILARION-Abastos	18
	8 Rattlesnake Canyon Grocery	18
Order	9 Bon app'	17
	10 Frankenversand	15

5. Revenue by product category

```
SELECT
    c.CategoryName,
    SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalRevenue
FROM Categories c
JOIN Products p ON c.CategoryID = p.CategoryID
JOIN [Order Details] od ON p.ProductID = od.ProductID
GROUP BY c.CategoryName
ORDER BY TotalRevenue DESC;
```

Categories(+) 1 X

SELECT c.CategoryName, SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS TotalRevenue

Grid	AZ CategoryName	123 TotalRevenue
Text	1 Beverages	267,868.1805229187
	2 Dairy Products	234,507.2852172852
	3 Confections	167,357.2248315811
	4 Meat/Poultry	163,022.3590888977
	5 Seafood	131,261.7374248505
	6 Condiments	106,047.0849895477
	7 Produce	99,984.5800685883
	8 Grains/Cereals	95,744.587474823